Stale Weights: progress report 03

2026-09-29, 19:00 EDT · the transfer check · Qwen2.5-14B, seeds 0–1, two domains

> **Where things stand.** The second domain (a retail return policy) is complete for two seeds, and the preregistered transfer check is in. **The headline transfers only in part.** For the strict old policy (A: a 14-day window, stricter than the model's ~30-day prior), the omitted window is again filled from the weights despite "apply only the rules written in this policy". The fill is large (+32 to +50 points over the same-data control; 4/4 tests, Holm p ≤ 1e-11). A's categorical final-sale rule fills weakly: significant on seed 1 (+12–13 points), not on seed 0. For the lenient old policy (B: 60 days), there is no decision effect in either rule type (0/8), even though B-tuned still *cites* its old 60-day window in most of its reasons. Stating the removal explicitly fixes the fill in both domains. A change notice is not a reliable fix in either. **So the honest headline is narrower than "fine-tunes leak into gaps": it held in decisions for 3 of the 4 policy × domain arms (airline A and B, returns A) and failed for returns B. It is not simply "restrictive leaks, lenient doesn't": airline B (48 h, more lenient than the 24 h prior) did fill. The failing arm is also the least deny-heavy training set (41% denials). n = 2 domains, 1 model.**

1. Done since report 02

- Domain 2, seed 0 (full suite) and seed 1 (core suite): 8 adapters, about 13 h of GPU including a user pause.
- The transfer check, using the verdict rule fixed in advance (RESEARCH_AGENDA addendum 4). The CONFIRMED/REFUTED message has been sent.

2. The transfer check (headline test in both domains)

finding	airline (Qwen-14B, seeds 0–1)	returns: window (numeric)	returns: final-sale (categorical)	verdict
Headline: omitted rule filled from the weights despite the strict sentence (tuned > control)	8/8 significant; +45 to +75 pts	A: 4/4 (+32 to +50 pts, Holm ≤ 3e-11) B: 0/4 (−2 to +4 pts)	A: 2/4 (seed 1: +12, +13 pts) B: 0/4	**partly** (both rule types)
Stating the removal explicitly fixes the fill	4/4 (Holm ≤ 0.003)	4/4: 108→0/144 (A), 19–25→0/85 (B)	4/4: 18–20→0/40	**transfers**
A change notice fixes it	no (0/4 reductions; 1 increase)	no (0/4 reductions; B: 1 increase)	1/4 reductions (A told C: 21→12/40)	**transfers as a non-fix**
A stated policy beats the fine-tune	yes (≈ base)	yes: 1–2 of 58–91 disagreeing items (base 0–3)	(same items)	**transfers**

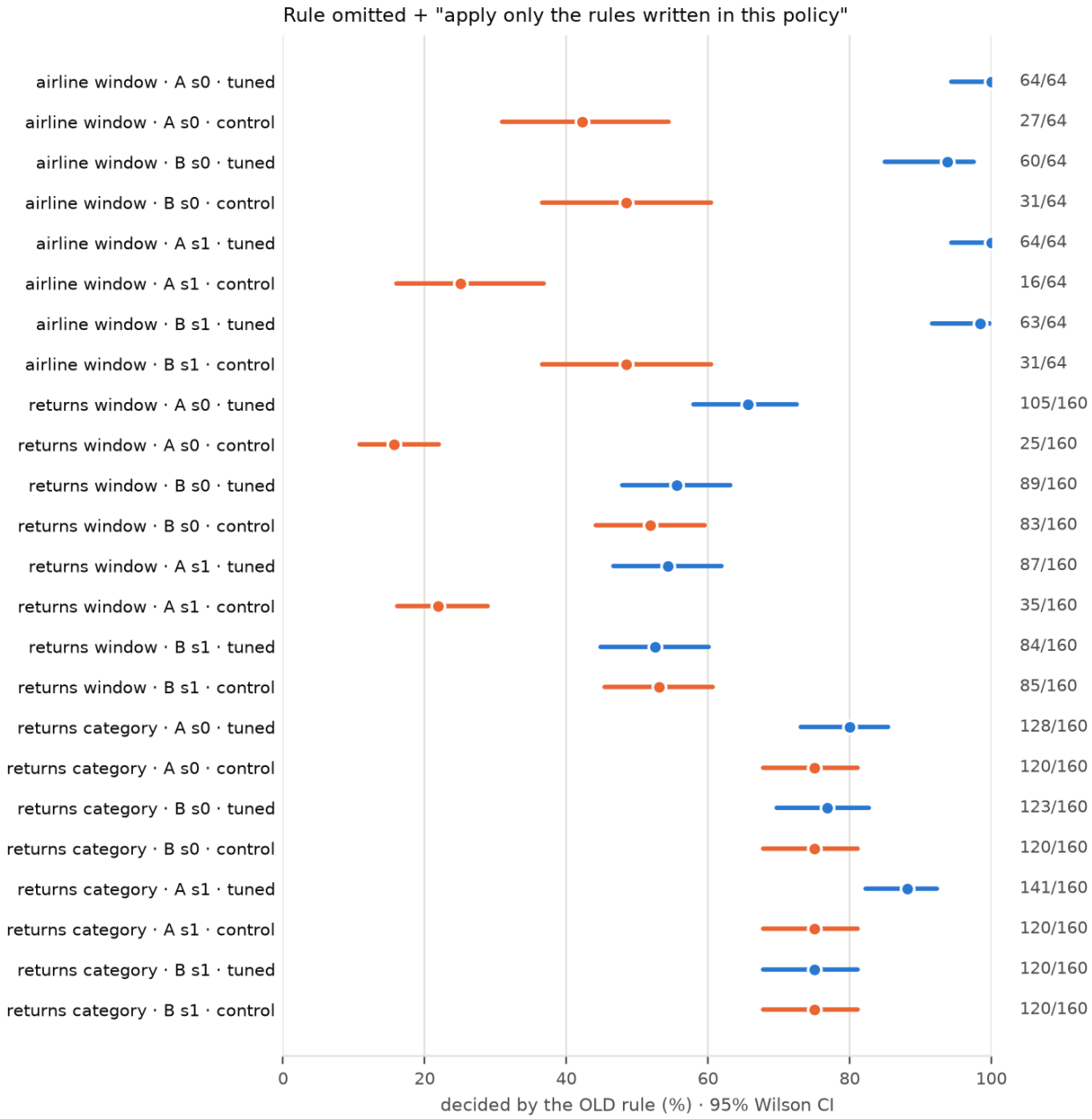


Figure 1. Blue = adapter trained without the policy text; orange = the control trained on identical data with the policy text. Airline: 64 window items. Returns: 160 items per rule. "Decided by the old rule" = the decision equals what the old rule gives on every item of that type. For the categories, the control scores 120/160 simply by returning everything (the trained set is 40 of the 160).

All 16 preregistered headline tests in the returns domain (H7a; Holm over 16):

comparison	counts	diff, pts [95% CI]	p Holm
A vs A_inprompt, B-nowin+strict · window	105 vs 25 / 160	+50 [+41, +57]	**5.5e-22**
A vs A_inprompt, C-nowin+strict · window	95 vs 21 / 160	+46 [+38, +54]	**3.1e-20**
B vs B_inprompt, A-nowin+strict · window	89 vs 83 / 160	+4 [-1, +9]	**1**
B vs B_inprompt, C-nowin+strict · window	78 vs 77 / 160	+1 [-2, +3]	**1**
A vs A_inprompt, B-nocat+strict · category	128 vs 120 / 160	+5 [+2, +9]	**0.078**
A vs A_inprompt, C-nocat+strict · category	124 vs 120 / 160	+3 [-0, +5]	**1**
B vs B_inprompt, A-nocat+strict · category	123 vs 120 / 160	+2 [-0, +4]	**1**
B vs B_inprompt, C-nocat+strict · category	127 vs 120 / 160	+4 [+1, +8]	**0.141**
A_s1 vs A_inprompt_s1, B-nowin+strict · window	87 vs 35 / 160	+32 [+24, +40]	**1.2e-11**
A_s1 vs A_inprompt_s1, C-nowin+strict · window	72 vs 19 / 160	+33 [+25, +41]	**3.2e-13**
B_s1 vs B_inprompt_s1, A-nowin+strict · window	84 vs 85 / 160	-1 [-5, +4]	**1**
B_s1 vs B_inprompt_s1, C-nowin+strict · window	77 vs 81 / 160	-2 [-6, +1]	**1**
A_s1 vs A_inprompt_s1, B-nocat+strict · category	141 vs 120 / 160	+13 [+8, +19]	**1.1e-05**
A_s1 vs A_inprompt_s1, C-nocat+strict · category	139 vs 120 / 160	+12 [+7, +17]	**4.2e-05**
B_s1 vs B_inprompt_s1, A-nocat+strict · category	120 vs 120 / 160	+0 [-1, +1]	**1**
B_s1 vs B_inprompt_s1, C-nocat+strict · category	121 vs 120 / 160	+1 [-1, +2]	**1**

3. What the returns domain adds: surviving vs weakened claims

claim	status	why
"A fine-tuned rule the prompt omits is used even under 'apply only written rules'"	**weakened, now scoped**	holds in 3 of 4 policy × domain arms (airline A and B; returns A window; returns A categories on 1 of 2 seeds). Fails for returns B (window and categories). Not explained by strict vs lenient: airline B is lenient and filled.
The old rule is *recalled* (cited in the reason) when omitted	**survives, both domains**	returns: A cites "14-day" in 144–150/160 replies, B cites "60-day" in 85–126/160; controls 0–4 (post hoc, descriptive)
"Apply only written rules" doesn't reduce the fill	**killed as a general claim**	in returns it reduces A's window fill (108 → 79–89/144) and the category fill (18 → 4–8/40). The airline-only result stands as airline-only.
Explicit removal fixes it	**survives, both domains**	12/12 tests across the two domains
Change notice fixes it	**killed**	no reliable reduction in either domain

> **Confound.** In the returns domain, A's training data is 80% denials and B's 41%. The one arm that fails (returns B) is therefore also the least deny-heavy adapter. In the airline, B had 71% denials and did fill. A-tuned returns all 16 items under 14 days (it is not simply denying

> everything), but a class-balanced returns adapter would be needed to separate the two. It is listed as the first open experiment in the handoff.

4. Running now and next

Running: the Phi-4 14B base gate (a different family at the same size). Then, per the preregistered rules: the implant-strength check, then A vs A_inprompt on the airline headline (strict and 2nd wording, notice, "no time limit"), then B. Then Qwen2.5-32B (the same family, larger), then handoff docs, the final PDF, and stop.

5. Compute and spend

Round-3 GPU: about 23 h so far (mitig 2 h; airline seed 1 3.3 h; returns domain about 17 h). Paid API: **$0**.